AI Engineer Take-Home Assignment

Financial Reconciliation System Using Machine Learning

Your task is to build a system that automatically matches transactions between two independent financial data sources (e.g., bank statements and internal check registers) using unsupervised machine learning techniques, drawing inspiration from a research paper.

Background Context

Financial reconciliation is the process of cross-checking independent sources of data to ensure completeness, timeliness, and accuracy of financial transactions. The research paper you've been provided demonstrates how this problem can be reframed as a "translation" problem between two different "languages" (different data sources describing the same transactions).

The Dataset

We've provided you with two CSV files containing synthetic financial data:

1. bank_statements.csv (308 transactions)

   - Columns: transaction_id, date, description, amount, type, balance

2. check_register.csv (308 transactions)

   - Columns: transaction_id, date, description, amount, type, category, notes

About the data:

- Both files describe the SAME 308 underlying transactions, but with realistic differences

- Dates may vary by 0-5 days (check register is typically earlier)

- Descriptions use different wording and formatting

- Transaction types use different codes (DEBIT/CREDIT vs DR/CR)

- Amounts occasionally have small rounding differences

- Some transactions have non-unique amounts (making matching harder)

- A few transactions are intentionally "anomalous" to test robustness

Your goal is to automatically match each bank transaction to its corresponding check register transaction.

This data has been generated synthetically using AI tools, feel free to modify or extend it.

Core Implementation

Implement a reconciliation system with the following components:

2.1 Unique Amount Matching

Implement automatic matching for transactions where the amount is unique across the entire dataset.

Requirements:

- Identify all transaction pairs where amounts uniquely match
- Calculate a confidence score for each match
- Flag potential issues (e.g., date differences exceeding expected thresholds)

2.2 Machine Learning-Based Matching

This is the core of the assignment. Implement an unsupervised learning approach for matching non-unique transactions.

You have flexibility in your approach, but consider these techniques:

Option A - SVD-based approach (as in the paper):

- Build term-alignment matrices based on transaction descriptors
- Calculate mutual information between descriptor terms
- Create term-by-transaction matrices
- Apply SVD for dimensionality reduction
- Use cosine similarity in the reduced space to find nearest neighbors

Option B - Modern embedding-based approach:

- Use pre-trained language models (BERT, sentence transformers, etc.) to create embeddings of transaction descriptions
- Combine textual embeddings with numerical features (amounts, dates)
- Use similarity metrics to find best matches
- Consider techniques like approximate nearest neighbor search

Option C - Custom hybrid approach:

- Propose and implement your own approach combining multiple techniques
- Must be data-driven and not purely rule-based

Core Requirements (regardless of approach):

- Learn from previously matched transaction pairs (simulate having some "ground truth" matches as training data)
- Handle variations in dates, descriptions, and transaction types
- Produce a confidence score for each proposed match
- The system should improve as more validated matches are added to the training set

2.3 Evaluation Metrics

Implement the following metrics to evaluate your system's performance:

```
Precision = |{correctly matched}| / |{all system matches}|
Recall = |{correctly matched}| / |{all transactions that should
match}|
F1 Score = 2 × (Precision × Recall) / (Precision + Recall)
```

Deliverable:

- Working code implementing the workflow
- Can be command-line based or simple web interface
- Should demonstrate the full cycle: match → review → improve

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Part 3: Analysis & Documentation

Provide a comprehensive analysis of your implementation:

Performance Analysis:

- Report precision, recall, and F1 scores
- Analyze which types of transactions are hardest to match
- Show how performance improves with more training data

Design Decisions:

- Explain your choice of ML approach
- Justify any departures from the paper's methodology
- Discuss trade-offs you considered

Limitations & Future Improvements:

- What are the weaknesses of your current implementation?
- What would you do with more time?
- How would you handle edge cases?

Technical Specifications

Programming Language:

- Use any language you're comfortable with (Python, JavaScript, Java, etc.)

Libraries & Frameworks:

- You're free to use any open-source libraries
- Common choices: NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, sentence-transformers

Code Quality:

- Write clean, well-documented code
- Include docstrings/comments for complex logic
- Follow language-specific style guides
- Include unit tests for critical functions (bonus points)

Version Control:

- Submit via Git repository (GitHub, GitLab, etc.)
- Include a .gitignore file
- Commit history should show your development process

Time Expectation

This assignment is designed to take approximately **8 hours** for a qualified candidate. You have **one week** from receipt to submit.

Submission Guidelines

1. **Submit via email to:** ankush.abrol@osfin.ai
2. **Include:**
    - Link to your Git repository (make sure it's public or add us as collaborators)
    - Brief cover note explaining your approach (2-3 paragraphs)
    - Any questions or clarifications you'd like to discuss
3. **Subject Line:** "AI Engineer Assignment - [Your Name]"

Research Paper Reference

- **Citation:** Peter A. Chew. 2020. Unsupervised-Learning Financial Reconciliation: a Robust, Accurate Approach Inspired by Machine Translation. ICAIF '20, October 15–16, 2020, New York, NY, USA.

You're encouraged to read the paper thoroughly, but you don't need to implement every detail exactly as described. Use it as inspiration and adapt the concepts to create a practical, working system. **Good luck! We're excited to see your solution.**